

e-CAM121_CUONX

eCAM_argus_camera Build and Install Guide_User_Manual



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e-con Systems

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Introduction to eCAM_argus_camera

eCAM_argus_camera is a sample application provided by e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras to use NVIDIA® libargus application programming interface (API).

eCAM_argus_camera application is based on argus_camera which is NVIDIA® default camera application in Jetson Orin Nano™/Orin™ NX development kit. It is a modified video viewer and capture software for the camera in the Jetson Orin Nano™/Orin™ NX development kit and demonstrates the features of e-CAM121_CUONX.

The command in this document is represented by color as shown in the following table.

Table 1: Notation of Color

Color	Notation
Blue	Commands running in development kit

This document explains how to install eCAM_argus_camera application executable files from the delivery package and how to build and install the eCAM_argus_camera application from the source.

Prerequisites

The libraries such as cmake, build-essential, gtk and so on are required to build eCAM_argus_camera software package. Please refer to the *e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf* for the compatible Linux distribution version (L4T version).

The package requirements are as follows:

- cmake
- build-essential
- pkg-config
- X11
- gtk+-3.0>=3.0.0
- expat
- JPEG
- gstreamer-1.0
- v4l-utils
- libv4l2-dev

Note: You must make sure that the Jetson Orin Nano™ development kit contains all the dependencies.

Installing Build Dependencies

The steps to install the build dependencies are as follows:

1. Run the following commands in the Jetson Orin Nano™/Orin™ NX development kit to enable all the repositories which are required for installing the dependencies.

```
sudo apt-add-repository universe  
sudo apt-get update
```

Note: Make sure that you have connected the Jetson Orin Nano™ development kit to a stable network.

2. Run the following command to install the dependencies in the Jetson Orin Nano™/Orin™ NX development kit.

```
sudo apt-get install cmake build-essential pkg-config libx11-dev  
libgtk-3-dev libexpat1-dev libjpeg-dev libgstreamer1.0-dev v4l-  
utils libv4l-dev cuda-toolkit-11-4
```

Description

The eCAM_argus_camera application is a simple interface for capturing and viewing video from the devices supported on the Jetson Orin Nano™/Orin™ NX development kit.

Using eCAM_argus_camera application, you can perform the following:

- Change resolution, frame rate and bits per pixel (bpp), if different resolutions are supported by the device.
- Display the currently configured values of preview.
- Capture the still images and set the path where still images will be saved.
- Select the required frame rate.

All the above listed properties can be configured by attractive and easy to use Graphical User Interface (GUI).

Identifying Deliverables

This section describes about identifying the deliverables.

The release package contains the application source code, eCAM_argus_camera application executable and documents. Please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf for the compatible Linux distribution version.

The steps for identifying the deliverables are as follows:

1. Copy the release package tar file to the home directory of the board.
2. Run the following commands to extract the e-CAM121_CUONX release package.

```
tar -xf e-  
CAM121_CUONX_<L4T_version>_<release_date>_<release_version>.tar.gz  
  
cd e-  
CAM121_CUONX_<L4T_version>_<release_date>_<release_version>
```

The source code for the eCAM_argus_camera application is present in the release package in the **Application/eCAM_argus_camera/Source/eCAM_argus_camera.tar.bz2** location.

The eCAM_argus_camera application executables for aarch64 is present in the release package inside this **Application/Binaries/eCAM_argus_camera/** location.

Please refer to the *Building and Installing eCAM_argus_camera from Source* section to build application from the source or refer to the *Installing Application Executables* section to use eCAM_argus_camera application.

Building and Installing eCAM_argus_camera from Source

This section describes about building and installing eCAM_argus_camera from the source.

The steps to build and install eCAM_argus_camera from the source are as follows:

1. Run the following commands to navigate to the application source directory in the Orin Nano™/Orin™ NX development kit.

```
cd Application/eCAM_argus_camera/Source/  
tar -xvf eCAM_argus_camera.tar.gz  
cd eCAM_argus_camera/argus
```

2. Run the following commands to create the makefiles.

```
mkdir build && cd build  
cmake -DMULTI_SESSION_SUPPORTED=1 ..
```

If CMake cannot find an include path for any dependencies, it may be required to provide them explicitly. To include path for any dependencies, run the following commands.

```
'cmake -DOPENGLES_INCLUDE_DIR=/path/to/khronos/include  
s ..'
```

3. Run the following make command to build the eCAM_argus_camera application.

```
make eCAM_argus_camera -j4
```

The application will be built inside this **/argus/build/apps/camera/ui/camera/** location.

4. Run the following make install command to install the built application.

```
sudo make install
```

The eCAM_argus_camera application will be installed in **/usr/local/bin** location of Orin Nano™/Orin™ NX development kit. This application is used to capture and view video from the camera.

Note: Run the following command to achieve stable frame rate before launching the eCAM_argus_camera application in the Orin Nano™/Orin™ NX development kit.

```
sudo ~/max-isp-vi-clks.sh
```

Installing Application Executables

This section describes about installing the eCAM_argus_camera application executables.

The steps to install prebuilt files are as follows:

1. Run the following commands to install the application binaries in the Orin Nano™/Orin™ NX development kit.

```
cd e-CAM121_CUONX_<L4T_version>_<JP_version>_JETSON-ONX-ONANO_<release_version>/ONX_ONANO/Application/Binaries/eCAM_argus_camera/  
sudo cp eCAM_argus_camera /usr/local/bin/eCAM_argus_camera
```

The prebuilt files will be installed to **/usr/local/bin** directory on the Orin Nano™/Orin™ NX development kit.

Note: Make sure that the required dependencies listed in *Prerequisites* section are installed. If the required dependencies are not installed, the eCAM_argus_camera executable will not work properly.

2. Run the following command to achieve stable frame rate before launching the eCAM_argus_camera application in the Orin Nano™/Orin™ NX development kit.

```
sudo ~/max-isp-vi-clks.sh
```

3. Run the following command to launch the camera application.

```
$ eCAM_argus_camera
```

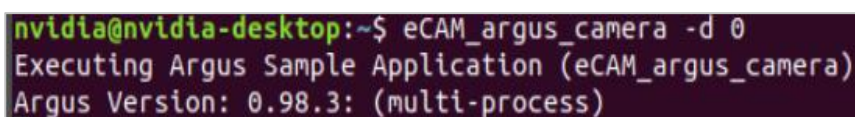
Multiple Cameras Streaming:

Run the following command to run the multiple cameras simultaneously.

```
$ eCAM_argus_camera -d n
```

Note: 'n' denotes the device number, where n will be 0-3 for quad cameras.

The following figure shows the execution of argus sample application for streaming multiple cameras.



```
nvidia@nvidia-desktop:~$ eCAM_argus_camera -d 0  
Executing Argus Sample Application (eCAM_argus_camera)  
Argus Version: 0.98.3: (multi-process)
```


Figure 1: Executing Argus Sample Application for Streaming Multiple Cameras

Troubleshooting

In this section, you can view the commonly occurring issues and their troubleshooting steps.

What can I do when I encounter error *in make or make install stage?*

Make sure that you have installed all the dependency packages listed in the *Prerequisites* section. Please refer to the [Installing Build Dependencies](#) section to install build dependencies and try again.

1. Is the eCAM_argus_camera application compatible to all the L4T versions?

The application is tested and verified in specific L4T versions. The steps provided in this document are not applicable to all the L4T/Jetpack™ versions. Please refer to the *e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf* for the compatible Linux distribution version (L4T version).

2. After fresh installation of quick start package, do I need to follow this document to setup eCAM_argus_camera application?

No, after fresh installation of quick start package, the application binary will be available in the Jetson Orin Nano™/Orin™ NX development kit. If you modify the application source, then refer this document to build and install the new version of application.

What's Next?

After understanding the build and installation procedure of eCAM_argus_camera application, you can refer to the following documents to understand more about e-CAM121_CUONX.

- *e-CAM121_CUONX Release Notes*
- *e-CAM121_CUONX eCAM_Argus App User Manual*

API: Application Programming Interface.

GIMP: GNU Image Manipulation Program.

GNU: GNU's Not Unix.

GTK: GIMP Toolkit.

JPEG: Joint Photographic Experts Group.

L4T: Linux for Tegra.

V4L2: Video for Linux version2.

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website -

<https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - [https://www.e-](https://www.e-consystems.com/create-ticket.asp)

[consystems.com/create-ticket.asp](https://www.e-consystems.com/create-ticket.asp)

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - [https://www.e-](https://www.e-consystems.com/RMA-Policy.asp)

[consystems.com/RMA-Policy.asp](https://www.e-consystems.com/RMA-Policy.asp)

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - [https://www.e-](https://www.e-consystems.com/warranty.asp)

[consystems.com/warranty.asp](https://www.e-consystems.com/warranty.asp)

Revision History

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1.0	24-May-2023	Initial Draft	Product Marketing Team